3.10. Connectness II

Def 3.4.9. (Sec(X) & Tan(X))

We define such varieties by sec bundle & tangent bundle.

$X \subseteq \mathbb{P}^r$ smooth irreducible of dim n.

(consider incidence variety $\mathcal{L} = \{(x, L) \in \mathbb{P}^r \times G \mid x \in L\}$. $G = Grass(r+1, r-n)$

then $\mathcal{L}/G$ universal family $\mathcal{L} \hookrightarrow \mathbb{P}^r \times G$.

$X \hookrightarrow \mathbb{P}^r$ gives $TX \hookrightarrow T\mathbb{P}^r|_X$ $\quad 0 \to \mathcal{O} \to \mathcal{O}(1)^{r+1} \to T_{\mathbb{P}^r} \to 0$

then $TX(-1)$ lift to $\widehat{TX} \subseteq \mathcal{O}_X^{r+1}$ (rank $n+1$) giving $X \xrightarrow{f} G$

$$\begin{array}{ccc} \mathcal{L} & \xrightarrow{\pi_2} & \mathbb{P}^r \\ \downarrow \pi_1 & & \\ X & \xrightarrow{f} & G \end{array}$$

$Tan(X) := \pi_2(\pi_1^{-1}(f(X)))$ closed subvariety.)

Consider $Hilb_2 X =: X^{[2]}$ universal family $\Sigma \subseteq X \times X^{[2]} \subseteq \mathbb{P}^r \times X^{[2]}$

fibre constant Hilbert poly $\Rightarrow$ $\pi_1 : \Sigma \to X$ $\quad \pi_2 : \mathbb{P}^r \times X^{[2]} \to X^{[2]}$

consider evaluation map $H^0(\mathcal{O}_{\mathbb{P}^r}(1) \boxtimes \mathcal{O}_{X^{[2]}}) \otimes \mathcal{O}_{\mathbb{P}^r \times X^{[2]}} \twoheadrightarrow \mathcal{O}_\Sigma(1)$

$\pi_2^* \rightsquigarrow \mathcal{O}_{X^{[2]}}^{r+1} \to \Sigma$ $\quad V = \mathbb{C}^{r+1}$

where $\mathcal{E} = \pi_{2*}\mathcal{O}_\Sigma = \pi_{1*}\mathcal{O}_\Sigma$ locally free of rank 2.

the surj map gives $X^{[2]} \to Gr(r+1, 2) = \{\text{lines in } \mathbb{P}^r\}$

$$\begin{array}{ccc} \mathcal{U} & \xrightarrow{\tilde{f}} & \mathbb{P}^r \\ \downarrow p_1 & & \\ X^{[2]} & \xrightarrow{f} & Gr(r+1,2) \end{array}$$

then $Sec(X) = p_2(p_1^{-1}(f(X^{[2]}))$ irreducible

Hilbert–Chow map $X^{[2]} \xrightarrow{\varphi} X^{(2)} = X \times X / S_2$ $\quad \Delta^{[2]} := \varphi^{-1}(\Delta)$ $\quad Tan(X) = p_2(p_1^{-1}(f(\Delta^{[2]})))$

irreducible.

$\Delta = Im(X \to X \times X \to X^{(2)})$

prop. 3.4.5. (Dimension of $Sec(X)$ & $Tan(X)$)

Either $\dim Sec(X) = 2n+1$ $\dim Tan(X) = 2n$ or $Sec(X) = Tan(X)$

proof of $Tan(X) \neq Sec(X)$ it suffices to show $\dim Tan(X) = 2n$

(If $\dim Tan(X) = 2n$, then $2n+1 \leq \dim Sec(X) \leq \dim(p^{-1}(f(X^{(n)})) = \dim f(X^{(n)}) + 1 \leq 2n+1$.)

Suppose $\dim Tan(X) < 2n$, then $\exists$ linear space L of codim $< 2n$ s.t.

$L \cap X = \emptyset \Leftarrow L \cap Tan(X) = \emptyset$ but $L \cap Sec(X) \neq \emptyset$ $(\dim Sec(X) > \dim Tan(X))$

projection with center L gives $X \xrightarrow{\varphi} \mathbb{P}^{r'}$ $r' = \text{codim } L - 1$, $r' < 2n$

φ is unramified but not inj on closed pts $\Rightarrow$ contradiction. □

Def. 3.4.7. (Local Degree)

① $f: X \to Y$ finite surj morphism of irre varieties of dim n. Y smooth

then $(\Omega^1_{X/Y})_{\eta_x} = \Omega^1_{k(x)/k(y)} = 0$ since C perfect. $V := X \setminus \mathrm{Supp}\,\Omega^1_{X/Y}$

$X \setminus V$ called ramification locus, $f(X \setminus V)$ called branching locus.

② (If Y smooth). $f^{-1}(Y \setminus Bran(f)) \to Y \setminus Bran(f)$ is Etale, since

$\forall y \in Y \setminus Bran(f)$ $x \in X_y$ $\dim T_y Y = \dim Y$ $\dim X \leq \dim T_x X \leq \dim T_y Y$

then $\dim X = \dim_x X \Rightarrow x \in X^{reg}$ then $f^{-1}(Y \setminus Bran(f))$ smooth

Miracle Flatness $\Rightarrow f^{-1}(Y \setminus Bran(f)) \to Y \setminus Bran(f)$ flat, flat + unramified $\Rightarrow$ Etale.

If Y not smooth, this may not be true. consider $| \to \alpha$.

② Analytically $\forall y \exists U(y)$ s.t. $f^{-1}(U(y)) = \coprod_{f(x')=y} V(x')$ $V(x')$ disjoint.

$V(x') \to U(y)$ still finite (quasi-finite + closed)

$e_f(x') := \deg(V(x) \to U(f(x)))$ well-defined deg is defined on unbranching locus

($\mathrm{Bran}(f)$ closed proper in Y, then $U \setminus \mathrm{Bran}(f)$ dense in U, take $y_n \to y$

$f|_{V(x)}$ is covering map near $y_n \Rightarrow \deg$ is constant, thus independent of U, V)

Obviously $\deg f = \sum_{f(x)=y} e_f(x)$ $\forall y$. $e_f(x)$ is analytically. upper semicontinuous.

(for $V(x) \xrightarrow{f} U(f(x))$ $\forall x' \in V(x)$ $\deg f = \sum_{f(x_i)=f(x')} e_f(x_i) \geq e_f(x) \geq e_f(x')$)

③ Higher ramification locus $R_\ell(f) := \{x \in X \mid e_f(x) > \ell\}$. closed. $R_1(f) = \mathrm{Ram}(f)$

analytic top

Fact: R_ℓ is analytic set thus Zar closed

Theorem 3.48. (Ramification of Covering to Proj space)

① X irreducible proj variety of dim n, $f: X \to \mathbb{P}^n$ branched covering of degree d, then $\exists x \in X$ s.t. $e_f(x) \geq \min\{d, n+1\}$.

① $\forall \ell \geq d$ $R_\ell(f) = \emptyset$ ② $\mathrm{codim}_X Z \leq \dim X = n$.

② In general, $\mathrm{codim}_X R_\ell(f) \leq \ell$ if $\ell \leq \min\{d-1, n\}$.

take $\ell = \min\{d-1, n\}$ $R_\ell(f) = \{x \in X \mid e_f(x) \geq \min\{d, n+1\}\}$ of codim $\leq n \Rightarrow \dim \geq 0$

proof of $d=1$ trivial $\mathrm{codim}_X X = 0$. WMA $d \geq 2$ $\ell \geq 1$.

induction, if $n=1$, $\ell=1$ Theorem $\Leftrightarrow R_1(f) \neq \emptyset$

if $R_1(f) = \emptyset$ f is étale, but $\pi_1^{ét}(\mathbb{P}^1) = 0$ contradiction

Assume $n \geq 2$, $L \subseteq \mathbb{P}^n$ general hyperplane s.t. $f^{-1}(L) =: X'$ irreducible

∃ $V \subseteq \mathbb{P}^n$ dense open s.t. $f^{-1}(V) \to V$ étale, then take L s.t.

$L \in \mathbb{P}^n \setminus V$, then $\forall y \in L \cap V$ $f^{-1}(L)_y = X_y$ contains d pts.

Claim. For general L, $e_f(x) = e_f(x)$ $\forall x \in f^{-1}(L)$. $\operatorname{codim} R_f = \operatorname{codim} R'_f$

$\forall x \in f^{-1}(L)$, take $y_n \to f(x)$ (analytic) $y_n \in L \cap V$ then $e_f(x) = e_{f'}(x) = \lim \#\, f|^{-1}_{V(y_n)}(y)$

modify L s.t. L "intersect" X

since $f^{-1}(L) \to L$ finite, $\deg(f|_X) = d$. induction $\Rightarrow f': X \to L = \mathbb{P}^{n-1}$ Theorem ✓

then $\operatorname{codim} R_f = \operatorname{codim} R'_f \le f$ $\forall f \le \min\{n-1, d-1\}$

it suffices to prove $f = n$ case i.e. $R_n \neq \emptyset$ $(n-1 < d-1)$

i.e. $\exists s^* \in X$ s.t. $e_f(s^*) \ge n+1$.

$\dim R_{n-1} = \dim X - \operatorname{codim} R_{n-1} \ge n - (n-1) = 1$

take a irr comp S of R_{n-1}, consider $F = f \times f|_S : X \overset{\cup}{\underset{\parallel}{\times}} S \to \mathbb{P}^n \times \mathbb{P}^n$

X irr S irr $\Rightarrow X \times S$ irr $\dim V = n + \dim S \ge n+1$

F is finite, $\dim F(X \times S) \ge n+1 > n$ then by Fulton–Hansen connectedness

$F^{-1}(\Delta_{\mathbb{P}^n})$ is connected, now $\Delta_S \subseteq S \times S$ $\Delta_S \subsetneq F^{-1}(\Delta_{\mathbb{P}^n})$

$X \times_{\mathbb{P}^n} S \to X$ $\dim X \times_{\mathbb{P}^n} S = \dim S = \dim \Delta_S$ + Δ_S irreducible

finite ↓ ↓ $S \to \mathbb{P}^n$ then Δ_S is irr component of $F^{-1}(\Delta_{\mathbb{P}^n})$

If $\Delta_S = F^{-1}(\Delta_{\mathbb{P}^n})$ $f(x) = f(s) \Rightarrow x = s$ then $\forall s \in S$ $\#f^{-1}(f(s)) = 1$

$e_f(S) = d$ then $d \ge n+1 \Rightarrow S \subseteq R_n$ ∨

If $\Delta_S \subsetneq F^{-1}(\Delta_{\mathbb{P}^n})$ then ∃ component $T \neq \Delta_S$ $T \cap \Delta_S \neq \emptyset$

fix $(x_k, s_k) \in T \subseteq X \times S$ with $x_k \neq s_k$ $f(x_k) = f(s_k)$ $(x_k, s_k) \to (s^*, s^*) \in \Delta_S \cap T$

then $\lim x_k = \lim s_k = s^*$ $(T \setminus \Delta_S$ dense in $T)$

$s_k \in R_{n-1}$ $e_f(s_k) \ge n$ $e_f(x_k) \ge 1$ $\Longrightarrow$ (Remark 3.9.9.) $e_f(s^*) \ge e_f(s_k) + e_f(x_k) \ge n+1$.

$s^* \in R_n$ then n case is done. □

Remark. 3.9.9. (additivity property)

$f: X \to Y$ finite surj morphism, we use analytic topology.

Consider $\{x_k\}$ $\{z_k\}$ sequence of pts in X. $x_k \neq z_k$ $f(x_k) = f(z_k)$

Assume $\lim x_k = \lim z_k = x^* \in X$ then $e_f(x^*) \geq \lim (e_f(x_k) + e_f(z_k))$

proof. take $f|: V(x^*) \to U(f(x^*))$ then $\forall y \in U$ $e_f(x^*) = \sum_{w \in V(x^*) \cap f^{-1}(y)} e_f(w)$

WLOG $\{x_k\}, \{z_k\} \subseteq V(x^*)$, $y_k := f(x_k) = f(z_k) \in U$

$e_f(x^*) = \sum_{w \in V(x^*) \cap f^{-1}(y_k)} e_f(w) \geq e_f(z_k) + e_f(x_k)$, $\exists k \gg 0$ s.t. $e_f(x_k) + e_f(z_k) = \lim (e_f(x_k) + e_f(z_k))$ (≥)

then $e_f(x^*) \geq \lim (e_f(x_k) + e_f(z_k))$

Corollary 3.9.10.

① X irr proj variety of dim n and $f: X \to \mathbb{P}^n$ branching covering of degree d.

if $d \leq n+1$ $\exists$ subvariety $T \subseteq \mathbb{P}^n$ of dim $\geq n+1-d$ over which f bijective.

② X irr normal proj variety of dim n, $f: X \to \mathbb{P}^n$ branched covering

of degree $d \leq n$, then $\pi_1^{et}(X) = 0$.

A Boundary example is $E \subseteq \mathbb{P}^n = \mathbb{P}$ elliptic of deg $n+1$

(take deg $n+1$ line bundle $\mathcal{L}/E$ then $n \geq 2 \Rightarrow \mathcal{L}$ very ample.

$h^0(E, \mathcal{L}) = \deg \mathcal{L} + 1 - g = n+1$ then $\mathcal{L}$ gives $E \hookrightarrow \mathbb{P}^n$)

$X := \{(x, H) \mid x \in H\} \subseteq E \times \mathbb{P}^\vee$ $p_2: X \to \mathbb{P}^\vee$ is finite covering of deg $n+1$.

but $p_1: X \to E$ is $\mathbb{P}^{n-1}$ bundle. $e = \pi_1(\mathbb{P}^{n-1}) \to \pi_1(X) \to \pi_1(E) \to \pi_0(\mathbb{P}^{n-1}) = e$

$\pi_1(E) = \mathbb{Z} \oplus \mathbb{Z}$

proof ② If $Y \xrightarrow{f} X$ connected étale covering of degree ≥ 2

then Y is irreducible (normal) since X normal. apply Theorem 3.4.8.

$g: Y \to \mathbb{P}^n$ of degree $\geq 2d$. $d \leq \min\{2d-1, n\} \leq \min\{\deg g, n\}$

$\exists y \in Y$ $e_g(y) \geq d+1$, but Y/X étale $\Rightarrow e_g(y) = e_f(p(y)) \leq d$ contradiction!

* Zak's Theorem.

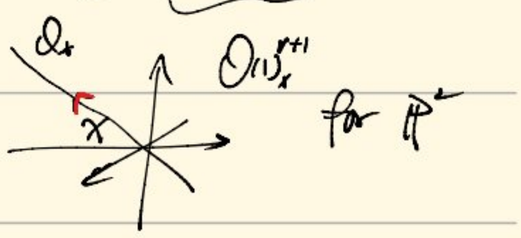
Goal: Smooth subvarieties of sufficiently small codim in proj space are linearly normal.

Notation: $X \subseteq \mathbb{P}^r$ smooth subvariety of dim n. $\forall x \in X$ $T_x \to \mathbb{P}^n$ is embedded tangent space

T_x is defined by

$$\begin{array}{ccc} T_x X & \hookrightarrow & T_x \mathbb{P}^r \\ \uparrow & \square & \uparrow \\ T_x & \to & \mathcal{O}(1)^{n+1}(x) \end{array}$$

$$0 \to \Omega_{\mathbb{P}^r} \to \mathcal{O}(-1)^{r+1} \to \mathcal{O} \to 0 \qquad 0 \to \mathcal{O} \to \mathcal{O}(1)^{r+1} \to T_{\mathbb{P}^r} \to 0$$



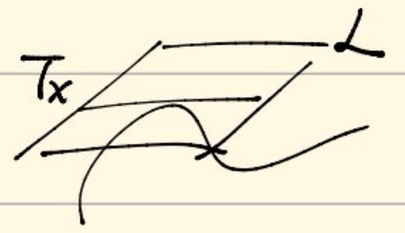
$0 \to Q_x(X) \hookrightarrow T_x \hookrightarrow \mathcal{O}(1)^{n+1}(x)$ T_x subspace of dim $n+1$

T_x containing $Q_x(X) \Rightarrow T_x \leftrightarrow$ linear subvariety of dim n passing x.

Def. 3.4.22 (Tangent)

① Now $\forall L$ linear subvariety, define L tangent to X at x iff

$x \in L \cap X$ and $T_x \subseteq L$



② $V \subseteq X$ subvariety of dim v.

then consider $X^{[2]} \to X^{(2)} = X \times X / S_2$ Hilbert-Chow morphism.

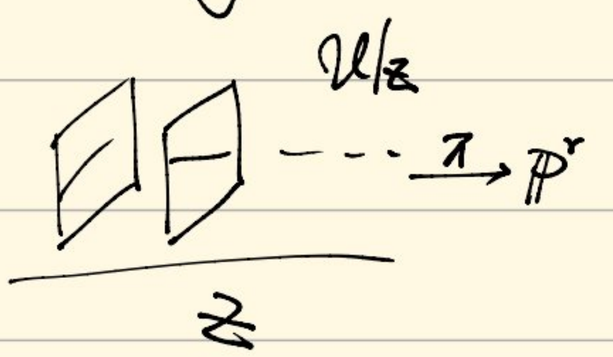
$$\begin{array}{ccccccc} V & \hookrightarrow & X \times V & \hookrightarrow & X \times X & \to & X \times X / S_2 \\ & \searrow & \downarrow & & \downarrow & & \uparrow \\ & & \mathcal{I}_V & \hookrightarrow & \mathcal{I}_\Delta & & \\ & & \uparrow & \square & \uparrow & & \\ & & Z' & \hookrightarrow & Z & \to & X^{(2)} = Bl_\Delta X \times X / S_2 \end{array}$$

$\dim N_{\Delta/X \times X} = n$

$\dim E_{X^{(2)}/X^{(1)}} = n+1$

$v+n$ $v+n$ $v+n+1$

then



$\mathrm{Sec}_v(X) := \pi(U/S)$ $\mathrm{Tan}_v(X) := \pi(U/S') \subseteq \mathrm{Sec}_v(X)$ $X \subseteq \mathrm{Sec}_v(X)$

Prop. 3.4.22.

Either $\dim \mathrm{Sec}_v(X) = n+v+1$ $\dim \mathrm{Tan}_v(X) = n+v$ or $\mathrm{Tan}_v(X) = \mathrm{Sec}_v(X)$.

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Proof Suppose $Tan_v(X) \neq Sec_v(X)$ Suppose $\dim Tan_v(X) = t < n+v$

A general linear space of codim $t+1$ meets $Sec_v(X)$ but not $Tan_v(X)$

(linear projection defines $X \xrightarrow{f} \mathbb{P}^t$, consider $V \times X \to E \to \mathbb{P}^t \times \mathbb{P}^v$

then $t < (n+v) = \dim(Im F) \Rightarrow F^{-1}(\Delta)$ Connected.

$\begin{cases} L \cap Sec_v(X) \neq \emptyset \Rightarrow \exists v_0 \in V \ x_0 \in X \ x_0 \neq v_0, f(x_0) = f(v_0) \text{ i.e. } (v_0, x_0) \in F^{-1}(\Delta)|_{\Delta_V} \\ L \cap Tan_v(X) = \emptyset \end{cases}$

take T another irre component of $F^{-1}(\Delta)$ s.t. $T \cap \Delta_V \neq \emptyset$

assume $(v^*, v^*) \in T \cap \Delta_V$, then $\exists (v_k, x_k) \in T|_{\Delta_V}$ $v_k, x_k \to v^*$ $k \to +\infty$

then $x_k \neq v_k$, $f(x_k) = f(v_k) \Rightarrow \overline{x_k v_k} \cap L \neq \emptyset$.

Consider

$$\begin{array}{ccc} \mathcal{U} & \longrightarrow & \mathbb{P}^n \\ \downarrow \pi & & \downarrow \\ Z & \hookleftarrow & L \end{array} \qquad \begin{array}{ccc} \mathcal{U} & \longrightarrow & \mathbb{P}^r \\ \uparrow & \square & \uparrow \\ \mathcal{U}_L & \longrightarrow & L \end{array}$$

$(V \times X|_{\Delta_V}) \to (V \times X|_{\Delta_V})/S_V \hookrightarrow Z$

π proper $\Rightarrow \pi(\mathcal{U}_L)$ closed

$(v_k, x_k) \in \pi(\mathcal{U}_k) \Rightarrow (v^*, v^*) \in \pi(\mathcal{U}_L)$ then $\exists l \in T_{v^*}$ $l \cap L \neq \emptyset$

which is a contradiction! □

Theorem, 3.4.1. (Zak's Thm)

$X \subseteq \mathbb{P}^r$ smooth irre subvar of dim n, assume X non-degenerate. Fix $L \subseteq \mathbb{P}^r$

of dim k with $n \leq k \leq r+1$, then $\{x \in X | T_x \subseteq L\} = \{x \in X | X \text{ tangent to } L \text{ at } x\}$

has dimension $\leq \min\{k-n, n-1\}$ (nondegenerate)

proof Suppose $\exists$ irre component $V \subseteq \{x \in X | T_x \subseteq L\}$ having dimension $\geq k+1-n$.

then $Tan_v(X) \subseteq L$ $\dim Tan_v(X) \leq \dim L = k \leq v+n-1$ since $v \geq k+1-n$.

by Prop. 3.4.22. $Tan_v(X) = Sec_v(X)$, but $X \subseteq Sec_v(X) = Tan_v(X) \subseteq L$

Contradiction!

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Corollary. 3.9 2018/12/20

① $X \subseteq \mathbb{P}^r$ irre sm. $G := Grass(r+1, r-n)$ then $f: X \to G$, $x \mapsto T_x$ finite.

② $H \subseteq \mathbb{P}^r$ hyperplane, then $\dim Sing(X \cap H) \leq r-1-n$

if $2n \geq r+1$ then $X \cap H$ reduced, if $2n \geq r+2$ then $X \cap H$ normal.

③ $P := \{(x, H) \mid T_x \subseteq H\} \subseteq X \times \mathbb{P}^r$ $\mathbb{P}^{r-n-1}$-bundle over X $\dim P = r-1$

then $p_2(P) =: X^*$ of dimension $\geq n$.

proof. ② Claim. $Sing(X \cap H) = \{x \in X \mid T_x \subseteq H\}$.

assume $H = V(s)$ $s \in H^0(\mathbb{P}^r, \mathcal{O}(1))$ then $s|_X \in H^0(X, \mathcal{O}(1))$

$X \cap H = V(s|_X)$, assume $s = \{U_i, f_i\}$. then

$\forall x \in U_i$ $X \cap H$ smooth at x iff $\partial x_n / f_i$ regular iff $\nabla \bar{f}_i \neq 0$

WMA $U_i = Spec A_i$ $X = V(I)$ $I = (s_{n+1}, \dots s_r)$ regular sequence (local chart)

extend to regular sequence $s_1, \dots s_r$ of $\mathbb{P}^r$

singular iff $\bar{f}_i \in m^2$ iff $\frac{\partial f_i}{\partial s_j} = 0$ $\forall 1 \leq j \leq r$. iff $T_x \subseteq V(f_i)$.

if $2n \geq r+1$ then $\dim Sing(X \cap H) \leq n-2 = \dim(X \cap H) - 1$, R_0

X smooth $\Rightarrow X$ CM $\Rightarrow X \cap H$ CM then $X \cap H$ S_1 (S_k $\forall k \geq 1$)

Fact: $R_0 + S_1 \Rightarrow$ reduced.

if $2n \geq r+2$ then $\dim Sing \leq \dim X \cap H - 2$ R_1

$R_1 + S_2 \Rightarrow$ normal.

③ $\forall H \in X^*$ fibre $p^{-1}(H) = \{x \in X \mid T_x \subseteq H\}$ $\dim \leq r-1-n$

then P / X^* flat $\Rightarrow \dim X^* \geq r-1-(r-1-n) = n$. □

Remark. 3.4.23. (Join Varieties)

Replace $Hilb_2X$ by $Bl_{\Delta_{X\times Y}} X\times Y/S_2$ then one can define $X*Y$ and $\mathcal{J}oin(X,Y)$

then either $\dim X*Y = n+m+1$ $\dim \mathcal{J}oin(X,Y) = n+m$. or $\mathcal{J}oin(X,Y) = X*Y$.

Theorem 3.4.25. (linear normality I)

$X\subseteq \mathbb{P}^r$ smooth proj variety of dim n, if $3n>2(r-1)$, then X is linearly normal

i.e. $H^0(\mathbb{P}^r,\mathcal{O}(1)) \xrightarrow{\varphi} H^0(X,\mathcal{O}_X(1))$ is surj.

since $X\subseteq \mathbb{P}\mathrm{Im}\varphi$ $3n>2(\dim \mathbb{P}\mathrm{Im}\varphi -1)$ WMA X nondegenerate.

Prop. 3.4.26

① X not linearly normal iff ② $\exists X\hookrightarrow \mathbb{P}^{r+1}$ s.t. $X\subset \mathbb{P}^{r+1}\dashrightarrow \mathbb{P}^r$

iff ③ $\exists X\hookrightarrow \mathbb{P}^{r+1}$ nondegenerate, $Sec(X)\subsetneq \mathbb{P}^{r+1}$, then

Theorem 3.4.25. $\Leftrightarrow$ $\forall X\subset \mathbb{P}^n$ sm non-degenerate variety of dim n

if $3n>2(m-2)$ then $Sec(X)=\mathbb{P}^m$ (linear normality II)

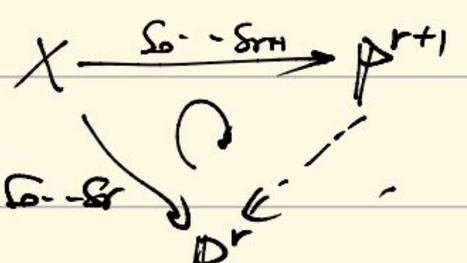
pf ① $\Leftrightarrow$ ②

X not linearly normal iff $\exists f\in H^0(X,\mathcal{O}_X(1))$ not in $\mathrm{Im}\varphi$.

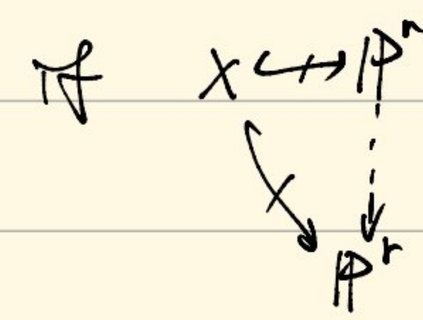
assume $H^0(\mathbb{P}^r,\mathcal{O}(1))\to H^0(X,\mathcal{O}(1))$, then $s_{r+1}:=f$, $s_0,\dots,s_{r+1}$ defines $X\to\mathbb{P}^{r+1}$

$x_0,\dots,x_r \mapsto s_0,\dots,s_r$

$X\to\mathbb{P}^{r+1}$ nondegenerated since $f\notin \mathrm{Im}\varphi$ and



conversely if $X\hookrightarrow\mathbb{P}^{r+1}$ then $X\hookrightarrow\mathbb{P}^r$ non-degenerate



and $H^0(\mathbb{P}^r,\mathcal{O}(1))\hookrightarrow H^0(\mathbb{P}^{r+1},\mathcal{O}(1))\hookrightarrow H^0(X,\mathcal{O}_X(1))$

② ⟺ ③ If $\exists$ $X \subset \mathbb{P}^{r+1} \dashrightarrow \mathbb{P}^r$ $P \notin Sec(X)$ (P center of projection)

Conversely if $Sec(X) \subsetneq \mathbb{P}^{r+1}$, take $P \notin Sec(X)$ □

Def 3.4.2 (Trisec)

$X \subseteq \mathbb{P}^m$ sm variety of dim n. We classify subvariety of dim 0 mult 3.

* If mult 2, $\dim \mathcal{O}_{X,x}/I = 2$ I sm since $\dim \mathcal{O}_{X,x}/\mathfrak{m} = 1 = \dim \mathcal{O}_{X,x}/\mathfrak{m}$

nakayama + $\bar{\mathfrak{m}} \neq 0$ $\bar{\mathfrak{m}}^2 \subsetneq \bar{\mathfrak{m}}$ $\dim \bar{\mathfrak{m}}/\bar{\mathfrak{m}}^2 \geq 1 \Rightarrow \dim \bar{\mathfrak{m}}/\bar{\mathfrak{m}}^2 = 1$ $\bar{\mathfrak{m}}^2 = 0$ since $\dim \bar{\mathfrak{m}}^2/\bar{\mathfrak{m}}^3 = 0$

$\bar{\mathfrak{m}}^2 = 0 \Rightarrow \mathfrak{m}^2 + I = I$ $\mathfrak{m}^2 \subseteq I$, take regular parameter $x_1 - x_n$, then

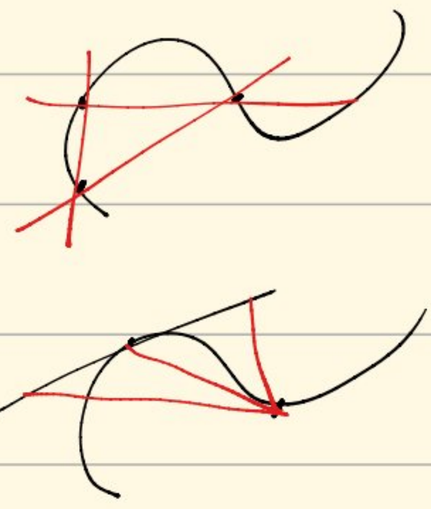
$\dim \bar{\mathfrak{m}}/\bar{\mathfrak{m}}^2 = \dim \bar{\mathfrak{m}} = \dim \mathfrak{m}/I = 1$ $\dim \mathfrak{m}/\mathfrak{m}^2 = n$, $\dim I/\mathfrak{m}^2 = n-1$, lift to I

$I = (x_i x_j, f_1 - f_{n-1})$ where $f_i \notin \mathfrak{m}^2$ $\operatorname{rank} \frac{\partial f_i}{\partial x_j} = n-1$

if J another ideal s.t. $J/\mathfrak{m}^2 = I/\mathfrak{m}^2 \subseteq \mathfrak{m}/\mathfrak{m}^2$, for lifting $g_1 - g_{n-1}$ to J

$f_i - \sum c_i g_i \in \mathfrak{m}^2 \Rightarrow f_i \in J$ $I \subseteq J$ $g_i \in I$ $J \subseteq I$, then $I = J$.

I is uniquely defined by tangent vector $(\mathfrak{m}/I)^\vee \hookrightarrow (\mathfrak{m}/\mathfrak{m}^2)^\vee$



For mult 3. first $A := \mathcal{O}/I = \begin{cases} A_1 \times A_2 \times A_3 & \dim A_i = 1 \\ A_1 \times A_2 & \dim A_1 = 1 \ \dim A_2 = 2 \\ A \ (\text{A}) \end{cases}$

$\dim A/\mathfrak{m} + \dim \mathfrak{m}/\bar{\mathfrak{m}}^2 + \dim \bar{\mathfrak{m}}^2/\bar{\mathfrak{m}}^3$ then $\mathfrak{m}^3 \subseteq I$.

$\quad 1 \quad\quad\quad \| \quad\quad\quad \| \quad\quad\quad \|$

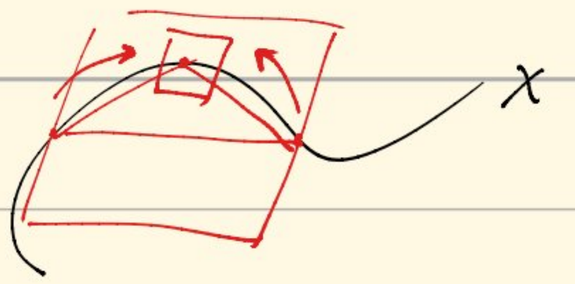
① if $\bar{\mathfrak{m}}^2 \neq 0$ $\dim \bar{\mathfrak{m}}/\bar{\mathfrak{m}}^2 = \dim \bar{\mathfrak{m}} = 1$ $\bar{\mathfrak{m}} = (t)$ $\bar{\mathfrak{m}}^2 = (t^2)$ $t^3 = 0$ $A = k[t]/t^3$

$\bar{\mathfrak{m}}/\bar{\mathfrak{m}}^2 = \mathfrak{m}/\mathfrak{m}^2 + I = k \cdot t$ $\mathfrak{m}^2 + I/\mathfrak{m}^2 = k t_2 \oplus k t_3 \cdots \oplus k t_n$ extend to $\mathfrak{m}/\mathfrak{m}^2 = \bigoplus_1^n k t_i$ $\bar{x}_i = t$

assume $x_1 \cdots x_n$ system of parameter, $\bar{x}_i = t_i$ $x_2 \cdots x_n \in I$, then $\bar{x}_1^3 = t^3 = 0$

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$I \supseteq (x_1^2, x_2 \cdots x_n) \Rightarrow I = (x_1^2, x_2 \cdots x_n)$



the plane π defined by $m/m^2 \to m/I$ $\quad (m/I)^\vee \hookrightarrow (m/m^2)^\vee = T_x^{(1)}X$ (dim 2)

$0 \to T_xX \to T_x^{(1)}X \to \mathrm{Sym}^2 T_xX \to 0$ $\quad \dim(n + \frac{n(n+1)}{2})$

$(0 \to m^2/m^3 \to m/m^3 \to m/m^2 \to 0)$ where $(m/I)^\vee \cap (m/m^2)^\vee$ of dim 1.

tangent space $H^0(\mathbb{P}^r, \mathcal{O}(1)) \twoheadrightarrow \mathcal{O}_{X,x}/m_x^2 \Rightarrow \mathbb{P}(\mathcal{O}_{X,x}/m_x^2)^\vee \hookrightarrow \mathbb{P}^r$

second tangent space $H^0(\mathbb{P}^r, \mathcal{O}(1)) \to \mathcal{O}_{X,x}/m_x^3 \Rightarrow \mathbb{P}(\mathcal{O}_{X,x}/m_x^3)^\vee \hookrightarrow \mathbb{P}^n$

$\uparrow$

$\mathbb{P}(\mathcal{O}_{X,x}/I)^\vee$

$\mathbb{P}(\mathcal{O}_{X,x}/I)^\vee$ may not contained in T_xX!

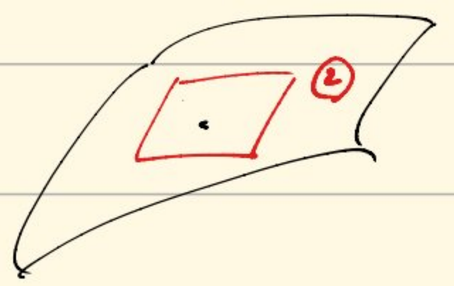
② If $\bar{m}^2 = 0$ $\dim \bar{m}/\bar{m}^2 = 2$. then $m^2 \subseteq I$ same argument

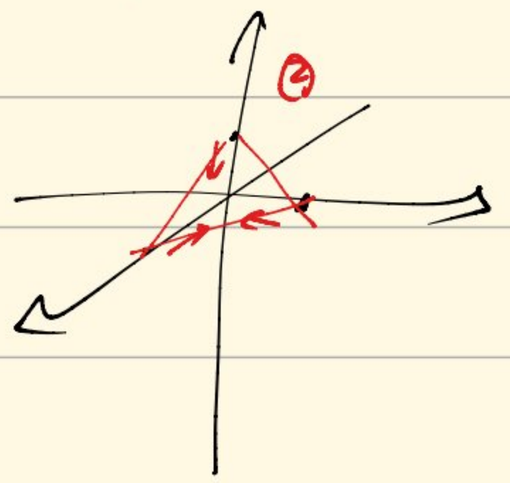
I is uniquely determined by m^2+I/m^2 of dim $n-2$

$\updownarrow$

$(m/m^2+I)^\vee \subseteq m/m^2{}^\vee$ dim 2 subspace of T_xX.

plane of this case already covered in $Sec(X)$





then one can define $Trisec(X) \supseteq Sec^3(X)$

by $\mathcal{U} \hookrightarrow \mathbb{P}^r \times Hilb_3(X) \to \mathbb{P}^n$ $\mathcal{U}$ family of planes.

$\downarrow$

$Hilb_3(X)$

or by $X * Sec(X) = X * X \supseteq X$.

Similarly $Sec^n(X) := \underbrace{X * X * X \cdots * X}_{n}$

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Remark. [Fogarty 68]

X sm variety, then $\mathrm{Hilb}_m(X)$ is smooth (iff) $\dim X \le 2$ or $m \le 3$.

$\mathrm{Hilb}_m X$ is irreducible if $\dim X \le 2$

$\mathrm{Hilb}_m(\mathbb{P}^n)$ is irreducible if $m \le 7$ [Maz08]

$\mathrm{Hilb}_8(\mathbb{P}^n)$ irr if $n \le 3$ & with 2 components if $n \ge 4$. [CEVV09]

$\mathrm{Hilb}_m(\mathbb{P}^3)$ for $m \ge 8$ remains unknown ($H_{11}(\mathbb{P}^3)$ irr [DJUNT17])

Lemma 3.4.28. (Terracini's Lemma)

Let $X \subseteq \mathbb{P}^m$ sm irr, then

① If $x, y \in X$ $x \neq y$, $p \in \overline{xy}$ then $\mathrm{Span}(T_x, T_y) \subseteq T_p \mathrm{Sec}(X)$

for general $x, y \in X$, $p \in \overline{xy} \cap \mathrm{Sec}(X)^{reg}$ $\mathrm{Span}(T_x, T_y) = T_p \mathrm{Sec}(X)$.

② If $x, y, z \in X$ noncollinear, $\ell \in \overline{xyz}$ plane, then $\mathrm{Span}(T_x, T_y, T_z) \subseteq T_\ell \mathrm{Trisec}(X)$

for general x, y, z, $p \in \overline{xyz} \cap \mathrm{TriSec}(X)^{reg}$ $\mathrm{Span}(T_x, T_y, T_z) = T_p \mathrm{TriSec}(X)$

proof omitted.

Corollary 3.4.29.

If $X \subseteq \mathbb{P}^m$ sm nondegenerated proj variety, then $\mathrm{Sec}(X) = \mathbb{P}^m$ iff $\mathrm{Sec}(X) = \mathrm{Trisec}(X)$

proof Since $\mathrm{Sec}(X) \subseteq \mathrm{Trisec}(X)$, it suffices to show if $\mathrm{Sec}(X) \neq \mathbb{P}^m$ $\mathrm{Sec} \neq \mathrm{Trisec}$

if $x, y \in X$ general pts T_x, T_y span $\Lambda \subseteq \mathbb{P}^m$

and $\Lambda = T_p \mathrm{Sec}(X)$ for general $p \in \mathrm{Sec}(X)$ by Lemma 3.4.28.

$(X \subsetneq \Lambda)$ choose $z \in X \setminus \Lambda$ then $\mathrm{Span}(T_x, T_y, T_z) \supsetneq \mathrm{Span}(T_x, T_y)$

then $\dim \mathrm{TriSec}(X) > \dim \mathrm{Sec}(X)$ □

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prop. 3.4.30. (proof of Theorem 3.4.25.)

$X \subseteq \mathbb{P}^n$ smooth proj variety of dimension n, if $3n > 2(n-2)$

then $Sec(X) = TriSec(X)$ i.e. $Sec(X) = \mathbb{P}^n$

(linear normality II) ✓ $\Longrightarrow$ (linear normality I) ✓

proof. Suppose $Sec(X) \subsetneq TriSec(X)$, then $\exists x_0, y_0, z_0 \in X$ (distinct) $\overline{x_0y_0z_0} \not\subseteq Sec(X)$

(if same point, adjust slightly then we still have $\overline{x_0y_0z_0} \not\subseteq Sec(X)$)

then $Sec(X) \cap \overline{xyz}$ proper closed subset $\Rightarrow$ dim ≤ 1

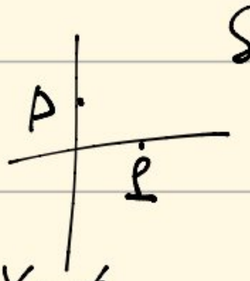
$\overline{xyz} = \mathbb{P}^2$

Now for general line $L \subseteq \overline{xyz}$ $L \cap Sec(X)$ (transversally) finite pts

Claim. $X \cap \overline{xyz}$ = finite points

If $\dim X \cap \overline{xyz} = 1$ $X \cap \overline{xyz}$ is union of line or $Sec(X) \cap \overline{xyz} = \overline{xyz}$

if not irreducible $Sec(X) \supseteq \overline{xyz}$ contradiction.



now adjust L s.t. $L \cap X = \emptyset$

then $\pi: X \longrightarrow \mathbb{P}^{n-2}$ by projecting from L, π finite morphism

consider $\pi \times \pi \times \pi =: F: X \times X \times X \longrightarrow \mathbb{P}^{n-2} \times \mathbb{P}^{n-2} \times \mathbb{P}^{n-2} \xleftarrow{\Delta} \mathbb{P}^{n-2}$

$F^{-1}(\Delta_{\mathbb{P}^{n-2}})$ connected by Fulton-Hansen. (3.3.8).

$\forall W$ irre comp. $\exists$ sequence $W_0 - - W_s$ s.t. $W_i \cap W_{i+1} \neq \emptyset$ $W_s \cap D_X \neq \emptyset$ $D_X := \Delta_X \times X \cup X \times \Delta_X$.

Now take W component containing (x_0, y_0, z_0)

one can find $\{(x_t, y_t, z_t)\}_{t \in T} \in X \times_{\mathbb{P}^{n-2}} X \times_{\mathbb{P}^{n-2}} X$ T smooth irr curve

such that $\exists! \, t^* \in T$, $(x^*, y^*, z^*) := (x_{t^*}, y_{t^*}, z_{t^*}) \in D_X$, $(x_1, y_1, z_1) \in W_s \cap W_{s-1}$

and $(x_t, y_t, z_t) \in W_s \setminus D_X$ for $T \setminus \{t^*\}$ (Recall

$\forall (x, y, z) \in F^{-1}(\Delta_{\mathbb{P}^{n-2}})$ $\pi(x_t) = \pi(y_t) = \pi(z_t)$ $\overline{xy}, \overline{yz}, \overline{xz} \cap L \neq \emptyset$

then define $\varphi: (x, y, z) \longmapsto (\overline{yz} \cap L, \overline{xz} \cap L, \overline{xy} \cap L)$ continuous

$F^{-1}(\Delta_{\mathbb{P}^{n-2}}) \longrightarrow L^3$

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since $L \cap Sec(X)$ finite, ϕ is locally constant

then $F^{-1}(\Delta_{\mathbb{P}^{n*}})$ connected $\Rightarrow$ ϕ constant, $\phi=(a,b,c)$

$a=\overline{x_t y_t} \cap L$ $b=\overline{x_t z_t} \cap L$ $c=\overline{y_t z_t} \cap L$ distinct and independent of t if $t \neq t^*$

(if $a=b$, then $\overline{x_0 y_0} \cap L = \overline{x_0 z_0} \cap L = a \Rightarrow x_0 = a = b \quad x_0 \in L$ x)

now if l^*_{xy} l^*_{xz} l^*_{yz} limits of secants as $t \mapsto t^*$ (DVR criterion)

$l^*_{xy} \cap L = a$ $l^*_{xz} \cap L = b$ $l^*_{yz} \cap L = c$ $\Rightarrow$ l^*_{ij} distinct.

if $y^* = z^* \neq x^*$, then $\overline{x_t y_t}$, $\overline{x_t z_t}$ degenerates to a common line λ

if $y^* = z^* = x^*$, then l^*_{ij} 3 distinct tangent line of x^*

$L \in T_{x^*} \Rightarrow L \subseteq Sec(X)$ contradiction. D

* Zariski's Problem.

Theorem. 3.9.33. (branching over nodal curve)

$C \subseteq \mathbb{P}^2$ possibly reducible curve with nodal singularity ($k[x,y]/(xy)$)

$f: X \to \mathbb{P}^2$ finite cover branched $B(f) \subseteq C$, then f Galois, $Gal(X/\mathbb{P}^2)$ abelian.